

Longitudinal magnetoeconductance of higher-pseudospin fermions

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Electron transport in Weyl semimetals has attracted significant attention in the condensed matter community due to the solid-state realization of Weyl fermions. While magnetotransport in pseudospin-1/2 (Weyl) systems is well established, its extension to higher-pseudospin fermions remains a frontier with critical implications for transport phenomena in materials with multifold fermions. We present a rigorous quasiclassical analysis of longitudinal magnetotransport in pseudospin-1 fermions, advancing beyond conventional models that assume constant relaxation times and neglect the orbital magnetic moment and global charge conservation. Our study uncovers a magnetic-field dependence of the longitudinal magnetoeconductance: it is positive and quadratic in B for weak internode scattering and transitions to negative values beyond a critical internode scattering strength. Notably, the critical threshold is lower for pseudospin-1 fermions compared to their pseudospin-1/2 counterparts. We show analytically that the zero-field conductivity is affected more strongly by internode scattering for pseudospin-1 fermions than conventional Weyl fermions. Furthermore, we extend our study to fourfold-degenerate fermions (pseudospin 3/2), which exhibit multiple Fermi surfaces, and uncover the nuanced nature of magnetotransport arising from distinct scattering channels. These insights provide a foundational framework for interpreting recent experiments on multifold fermions and offer a road map for studying magnetotransport in candidate materials with space group symmetries 199, 214, and 220.

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I. INTRODUCTION

The definitive discovery of Weyl fermions in solids has triggered intense research in the past decade [1–10]. Theoretical advances, however, date back to the 1980s when Nielsen and Ninomiya, who studied lattice Weyl fermions, proved that they must occur in pairs of opposite chiralities [11,12]. Such pairing ensures the conservation of both the chiral and global charge. When external gauge fields are present, chiral charge is not conserved, a phenomenon that is now well-known as the chiral anomaly or the Adler-Bell-Jackiw anomaly [13] of Weyl fermions. The manifestation of this anomaly is investigated through transport, thermoelectric, and optical experiments in systems hosting Weyl fermions, known as Weyl semimetals (WSMs) [14–27].

While Weyl fermions have made an entry from high-energy physics to condensed matter, certain symmetries of condensed matter systems make it possible to realize free fermionic excitations which are not allowed by Poincaré symmetry in high-energy physics [28–30]. Specifically, the low-energy Hamiltonian of a Weyl fermion is $H_{\mathbf{k}} \sim \chi \mathbf{k} \cdot \boldsymbol{\sigma}$, where $\boldsymbol{\sigma}$ is the vector of Pauli matrices, and χ is the chirality of the fermion. Fermionic excitations of the type $H_{\mathbf{k}} \sim \chi \mathbf{k} \cdot \mathbf{S}$ are allowed in solids, where $\mathbf{S}$ is a higher (pseudo)spin generalization of the vector of Pauli matrices. These excitations are multifold degenerate chiral quasiparticles, carry a nontrivial Chern number $|\mathcal{C}| > 1$, and are sources and sinks of the Berry curvature. In contrast, Weyl fermions have a twofold degeneracy with Chern number $|\mathcal{C}| = 1$.

Although Berry curvature induced magnetotransport effects have been heavily discussed in the context of Weyl

fermions or even Kramer-Weyl fermions [31–33], the generalization of such effects beyond fermions of pseudospin-1/2 has received little attention. On generic grounds, chiral particles respond differently to external magnetic fields, and one does expect similar effects to persist for higher pseudospins as well. In the context of pseudospin-1/2 Weyl fermions, when external fields are applied such that $\mathbf{E} \cdot \mathbf{B} \neq 0$, the chiral charge is not conserved and a chiral current is generated via the chiral anomaly (CA). Similar conclusions can be derived using quasiclassical approaches by accounting for Berry-curvature effects [18]. Recently, there have been theoretical advances based on this idea, aimed at understanding the associated transport features in multifold fermions [34–38]. Still, the quasiclassical analysis suffers from shortcomings such as imposing a constant relaxation time, neglecting orbital magnetic moment, internode scattering effects, and global charge conservation. Going beyond these standard assumptions is indispensable to correctly reproduce the physics of chiral anomaly and magnetotransport in chiral fermions in the experimentally accessible low-field limit as shown in some recent works [39–43].

This work presents a complete quasiclassical analysis of chiral anomaly and longitudinal magnetotransport in pseudospin-1 fermions. Moving beyond the conventional constant relaxation-time approximation and taking into account the effects of orbital magnetic moment and global charge conservation, we discover that chiral anomaly in pseudospin-1 fermions generates a positive and quadratic-in- B longitudinal magnetoeconductance for low internode scattering, which becomes negative as the strength of internode scattering (α) is increased beyond a critical value ($\alpha^{(1)}$).

Interestingly, this critical intervalley scattering strength is lower in pseudospin-1 fermions than the regular Weyl fermions ($\alpha^{(1)} < \alpha^{(1/2)}$). We also show analytically that the zero-field conductivity decreases sharply with internode scattering for pseudospin-1 fermions than conventional Weyl fermions. This study becomes even more pertinent in light of recent experiments that have probed this anomaly in multifold fermionic systems [44], and upcoming experiments that could be performed on relevant materials belonging to space groups 199, 214, and 220 [28]. We further extend our analysis to fourfold-degenerate pseudospin-3/2 fermions, which exhibit multiple Fermi surfaces and consequently support four distinct types of scattering processes: (i) internode interband, (ii) internode intraband, (iii) intranode interband, and (iv) intranode intraband. By isolating the effects of each scattering channel, we reveal the nuanced features they imprint on the magnetotransport behavior.

II. SEMIMETALS WITH THREEFOLD DEGENERACY

The low-energy Hamiltonian of the pseudospin-1 fermion expanded about a nodal point is [28]

$$H(\mathbf{k}) = \hbar v_F \mathbf{k} \cdot \mathbf{S}, \quad (1)$$

where v_F is a material-dependent velocity parameter, $\mathbf{k}$ is the wave vector measured from the nodal point, and $\mathbf{S}$ is the vector of spin-1 Pauli matrices. One may diagonalize this Hamiltonian to get energy eigenvalues:

$$\epsilon_k = 0, \quad \pm \hbar v_F k. \quad (2)$$

The dispersion consists of three bands, including a flat band with zero energy. The Bloch states corresponding to the nonzero energies are calculated to be

$$\begin{aligned} |u^+\rangle &= \left[\cos^2(\theta/2) e^{-2i\phi}, \frac{1}{\sqrt{2}} \sin(\theta) e^{-i\phi}, \sin^2(\theta/2) \right]^T, \\ |u^-\rangle &= \left[\sin^2(\theta/2) e^{-2i\phi}, \frac{-1}{\sqrt{2}} \sin(\theta) e^{-i\phi}, \cos^2(\theta/2) \right]^T, \end{aligned} \quad (3)$$

with θ and ϕ being the polar and azimuthal angles, respectively, i.e., $\mathbf{k} = k(\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$. The total Berry flux enclosing the Weyl points is now $\nu = \mp 2$ (modulo 2π), while the flat band is trivial. Such fermions may emerge in a body-centered cubic lattice with space groups 199, 214, and 220 at the $\mathbf{P}$ point of the Brillouin zone [28]. The $\mathbf{P}$ point does not map to its time-reversal partner ($\mathbf{P} \neq -\mathbf{P}$). Due to the Nielsen-Ninomiya theorem, the Brillouin zone must compensate for sources and sinks of the Berry curvature [11,12]. Therefore, the low-energy minimal Hamiltonian for pseudospin-1 semimetal may be written as

$$H(\mathbf{k}) = \sum_{\chi=\pm 1} \chi \hbar v_F \mathbf{k} \cdot \mathbf{S}. \quad (4)$$

The Berry curvature of the conduction bands of the Hamiltonian is evaluated to be

$$\Omega_{\mathbf{k}}^\chi = i \nabla_{\mathbf{k}} \times \langle u^\chi(\mathbf{k}) | \nabla_{\mathbf{k}} | u^\chi(\mathbf{k}) \rangle \equiv -\chi \mathbf{k} / k^3. \quad (5)$$

In contrast to a classical point particle, a Bloch wave packet in a crystal possesses a finite spatial extent. Consequently, it

undergoes self-rotation around its center of mass, resulting in an orbital magnetic moment (OMM), expressed as [45–47]

$$\begin{aligned} \mathbf{m}_{\mathbf{k}}^\chi &= -\frac{ie}{2\hbar} \text{Im} \langle \nabla_{\mathbf{k}} u^\chi | [\epsilon_0(\mathbf{k}) - \hat{H}^\chi(\mathbf{k})] | \nabla_{\mathbf{k}} u^\chi \rangle \\ &= -\chi e v_F \mathbf{k} / k^2. \end{aligned} \quad (6)$$

In the presence of an external magnetic field ($\mathbf{B}$), the orbital magnetic moment couples to it, and the dispersion relation is modified as $\epsilon^\chi(\mathbf{k}) \rightarrow \epsilon^\chi(\mathbf{k}) - \mathbf{m}^\chi(\mathbf{k}) \cdot \mathbf{B}$. Consequently, the Fermi contour becomes anisotropic:

$$k_F^\chi(\theta) = \frac{\epsilon_F + \sqrt{\epsilon_F^2 - \chi \eta \epsilon_0^2 \cos(\theta)}}{2 v_F \hbar}, \quad (7)$$

with ϵ_F being the Fermi energy, $\epsilon_0 = \sqrt{4eBv_F^2 \hbar}$, and the variable $\eta \in \{0, 1\}$ is used to toggle the effect of the orbital magnetic moment, such that its effect can be studied separately.

III. QUASICLASSICAL TRANSPORT

The dynamics of the quasiparticles in the presence of electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields, are described by the following equation [39,48]:

$$\begin{aligned} \dot{\mathbf{r}}^\chi &= \mathcal{D}_{\mathbf{k}}^\chi \left(\frac{e}{\hbar} (\mathbf{E} \times \boldsymbol{\Omega}^\chi) + \frac{e}{\hbar} (\mathbf{v}^\chi \cdot \boldsymbol{\Omega}^\chi) \mathbf{B} + \mathbf{v}_{\mathbf{k}}^\chi \right), \\ \dot{\mathbf{k}}^\chi &= -\frac{e}{\hbar} \mathcal{D}_{\mathbf{k}}^\chi \left(\mathbf{E} + \mathbf{v}_{\mathbf{k}}^\chi \times \mathbf{B} + \frac{e}{\hbar} (\mathbf{E} \cdot \mathbf{B}) \boldsymbol{\Omega}^\chi \right), \end{aligned} \quad (8)$$

where $\mathbf{v}_{\mathbf{k}}^\chi = \frac{1}{\hbar} \frac{\partial \epsilon^\chi(\mathbf{k})}{\partial \mathbf{k}}$ is the band velocity, $\boldsymbol{\Omega}^\chi = -\chi \mathbf{k} / 2k^3$ is the Berry curvature, and $\mathcal{D}^\chi = (1 + e\mathbf{B} \cdot \boldsymbol{\Omega}^\chi / \hbar)^{-1}$ is the factor by which the phase space volume is modified due to Berry phase effects [49]. We point out that we implicitly assume that the above semiclassical equations remain valid for pseudospin-3/2 fermions as well. Although the effective kinetic theory can be derived from a Landau-level basis for spin-1/2 fermions, such calculations have not yet been performed for spin-1 fermions. However, we emphasize that the above quasiclassical equations of motion should remain independent of the pseudospin magnitude as they arise purely from band geometry. Although a microscopic Landau level derivation for spin-1 fermions is lacking, recent studies support the validity of this framework for capturing low-field magnetotransport in such systems [44,50–52]. To describe the dynamics of three-dimensional pseudospin-1 fermions under external electric and magnetic fields, we employ the quasiclassical Boltzmann formalism, where the evolution of the nonequilibrium distribution function $f_{\mathbf{k}}^\chi$ is given by

$$\frac{\partial f_{\mathbf{k}}^\chi}{\partial t} + \dot{\mathbf{r}}_{\mathbf{k}}^\chi \cdot \nabla_{\mathbf{r}} f_{\mathbf{k}}^\chi + \dot{\mathbf{k}}_{\mathbf{k}}^\chi \cdot \nabla_{\mathbf{k}} f_{\mathbf{k}}^\chi = I_{\text{coll}}[f_{\mathbf{k}}^\chi], \quad (9)$$

with $f_{\mathbf{k}}^\chi = f_0 + g_{\mathbf{k}}^\chi$, where f_0 is the Fermi-Dirac distribution and $g_{\mathbf{k}}^\chi$ is the deviation. We fix the electric and magnetic fields along the z direction and the deviation up to the first order in perturbation is expressed as

$$g_{\mathbf{k}}^\chi = -e \left(\frac{\partial f_0}{\partial \epsilon} \right) \Lambda_{\mathbf{k}}^\chi E, \quad (10)$$

where $\Lambda_{\mathbf{k}}^{\chi}$ is unknown function to be evaluated. The collision integral considers two impurity-dominated distinct scattering processes: (i) scattering between χ and $\chi' \neq \chi$ and (ii) scattering between χ to $\chi' = \chi$. These are also known as internode (intervalley) scattering, and intranode (intravalley) scattering, respectively. The collision integral is expressed as [18,23]

$$I_{\text{coll}}[f_{\mathbf{k}}^{\chi}] = \sum_{\chi' \mathbf{k}'} \mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} (f_{\mathbf{k}'}^{\chi'} - f_{\mathbf{k}}^{\chi}), \quad (11)$$

where the scattering rate $\mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'}$ calculated using Fermi's golden rule:

$$\mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} = \frac{2\pi n}{\hbar \mathcal{V}} |\langle u^{\chi'}(\mathbf{k}') | U_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} | u^{\chi}(\mathbf{k}) \rangle|^2 \times \delta(\epsilon^{\chi'}(\mathbf{k}') - \epsilon_{\mathbf{k}}^{\chi}). \quad (12)$$

Here n represents the impurity concentration, $\mathcal{V}$ represents the system volume, and $U_{\mathbf{k}\mathbf{k}'}^{\chi\chi'}$ describes the scattering potential profile. If the energy is fixed to be the Fermi energy (ϵ_F), the scattering rate will then have only angular dependence. For elastic impurities, we set $U_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} = I_{3 \times 3} U^{\chi\chi'}$, where $U^{\chi\chi'}$ controls scattering between electrons of different ($\chi \neq \chi'$) and same chirality ($\chi = \chi'$). The relative magnitude of chirality-breaking to chirality-preserving scattering is expressed by the ratio

$$\alpha = \frac{U^{\chi\chi' \neq \chi}}{U^{\chi\chi' = \chi}}. \quad (13)$$

The overlap $\mathcal{T}^{\chi\chi'}(\theta, \theta', \phi, \phi') = |\langle u^{\chi'}(\mathbf{k}') | U_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} | u^{\chi}(\mathbf{k}) \rangle|^2$ is generally a function of both the polar and azimuthal angles, making the scattering strongly anisotropic. However, for the chosen arrangement of the fields, i.e., $\mathbf{E} = E\hat{z}$, and $\mathbf{B} = B\hat{z}$, the terms involving ϕ and ϕ' are irrelevant due to azimuthal symmetry as they vanish when the integral with respect to ϕ' is performed. We drop such terms and write the the overlap function as

$$\mathcal{T}_{\theta, \theta'}^{\chi\chi'} = \begin{cases} |U^{\chi\chi'}|^2 \mathcal{V}^2 [\cos^4(\theta/2) \cos^4(\theta'/2) \\ + \sin^4(\theta/2) \sin^4(\theta'/2) + \frac{1}{4} \sin^2(\theta) \sin^2(\theta')] \delta_{\chi, \chi'} \\ |U^{\chi\chi'}|^2 \mathcal{V}^2 [\sin^4(\theta/2) \cos^4(\theta'/2) \\ + \cos^4(\theta/2) \sin^4(\theta'/2) + \frac{1}{4} \sin^2(\theta) \sin^2(\theta')] \delta_{\chi, -\chi'}. \end{cases} \quad (14)$$

Using Eqs. (8), (10), and (11), Eq. (9) is written in the following form:

$$\mathcal{D}_{\mathbf{k}}^{\chi} \left[v_{\mathbf{k}}^{\chi, z} + \frac{eB}{\hbar} (\mathbf{v}_{\mathbf{k}}^{\chi} \cdot \boldsymbol{\Omega}_{\mathbf{k}}^{\chi}) \right] = \sum_{\chi' \mathbf{k}'} \mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} (\Lambda_{\mathbf{k}'}^{\chi'} - \Lambda_{\mathbf{k}}^{\chi}). \quad (15)$$

Note that we keep only terms up to linear order in the electric field, and furthermore terms involving the Lorentz force are not included as they vanish in the chosen geometry. Before further simplifying the above equation, we define the chiral scattering rate as follows:

$$\frac{1}{\tau^{\chi}(\theta)} = \sum_{\chi'} \mathcal{V} \int \frac{d^3 \mathbf{k}'}{(2\pi)^3} (\mathcal{D}_{\mathbf{k}'}^{\chi'})^{-1} \mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'}. \quad (16)$$

Note the additional factor of $\mathcal{D}_{\mathbf{k}}^{\chi}$ due to the modification of density of states as a result of Berry curvature [49]. The

volume factor on the right-hand side of the above equation drops out when the k -space integration is performed. Equation (15) then transforms to

$$h^{\chi}(\theta) + \frac{\Lambda^{\chi}(\theta)}{\tau^{\chi}(\theta)} = \sum_{\chi'} \mathcal{V} \int \frac{d^3 \mathbf{k}'}{(2\pi)^3} \mathcal{D}_{\mathbf{k}'}^{\chi'} \mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} \Lambda^{\chi'}(\theta'). \quad (17)$$

Here, $h^{\chi}(\theta) = \mathcal{D}_{\mathbf{k}}^{\chi} [v_{\mathbf{k}}^{\chi, z} + eB(\boldsymbol{\Omega}_{\mathbf{k}}^{\chi} \cdot \mathbf{v}_{\mathbf{k}}^{\chi})/\hbar]$, evaluated on the Fermi surface. Employing azimuthal symmetry, Eqs. (16) and (17) simplify to an integration over θ' :

$$\frac{1}{\tau^{\chi}(\theta)} = \sum_{\chi'} \Pi^{\chi\chi'} \int \frac{(k')^3 \sin \theta' d\theta'}{|\mathbf{v}_{\mathbf{k}'}^{\chi'} \cdot \mathbf{k}'^{\chi'}|} \mathcal{T}_{\theta\theta'}^{\chi\chi'} (\mathcal{D}_{\mathbf{k}'}^{\chi'})^{-1}, \quad (18)$$

$$h^{\chi}(\theta) + \frac{\Lambda^{\chi}(\theta)}{\tau^{\chi}(\theta)} = \sum_{\chi'} \Pi^{\chi\chi'} \int d\theta' f^{\chi'}(\theta') \mathcal{T}_{\theta\theta'}^{\chi\chi'} \Lambda^{\chi'}(\theta') / \tau^{\chi'}(\theta'), \quad (19)$$

where $\Pi^{\chi\chi'} = n/4\pi^2 \hbar^2$ (N being the number of impurities), $f^{\chi}(\theta) = \frac{(k)^3}{|\mathbf{v}_{\mathbf{k}}^{\chi} \cdot \mathbf{k}^{\chi}|} \sin \theta (\mathcal{D}_{\mathbf{k}}^{\chi})^{-1} \tau^{\chi}(\theta)$, and $\mathcal{T}_{\theta\theta'}^{\chi\chi'}$ is defined in Eq. (14). With the ansatz $\Lambda^{\chi}(\theta) = [\lambda^{\chi} - h^{\chi}(\theta) + a^{\chi} \cos^4(\theta/2) + b^{\chi} \sin^4(\theta/2) + c^{\chi} \sin^2(\theta)] \tau^{\chi}(\theta)$, Eq. (19) is expressed as

$$\begin{aligned} & \lambda^{\chi} + a^{\chi} \cos^4(\theta/2) + b^{\chi} \sin^4(\theta/2) + c^{\chi} \sin^2(\theta) \\ &= \sum_{\chi'} \Pi^{\chi\chi'} \int f^{\chi'}(\theta') d\theta' \mathcal{T}_{\theta\theta'}^{\chi\chi'} [\lambda^{\chi'} - h^{\chi'}(\theta') \\ &+ a^{\chi'} \cos^4(\theta'/2) + b^{\chi'} \sin^4(\theta'/2) + c^{\chi'} \sin^2(\theta')]. \end{aligned} \quad (20)$$

When explicitly written, this equation consists of eight simultaneous equations that need to be solved for eight variables, and particle number conservation serves as an additional constraint, given by

$$\sum_{\chi} \sum_{\mathbf{k}} g_{\mathbf{k}}^{\chi} = 0. \quad (21)$$

The current is calculated using the expression

$$\mathbf{J} = -e \sum_{\chi, \mathbf{k}} f_{\mathbf{k}}^{\chi} \dot{\mathbf{r}}^{\chi}. \quad (22)$$

IV. RESULTS AND DISCUSSION

A. LMC in pseudospin-1 fermions and comparison with spin-1/2 Weyl fermions

We study the longitudinal magnetoconductance of the pseudospin-1 semimetal subjected to parallel electric and magnetic fields ($\mathbf{E} \parallel \mathbf{B}$) in the semiclassical regime. This specific field arrangement allows us to restrict the current flow along the z direction, making it easy to make analytical progress. For parallel electric and magnetic fields, the Lorentz force ($\mathbf{F} \propto \mathbf{v} \times \mathbf{B}$) is zero, which makes no conventional transverse current possible in this orientation. Longitudinal magnetoconductance (LMC) is defined as

$$\delta\sigma_{zz}(B) = \sigma_{zz}(B) - \sigma_{zz}(0). \quad (23)$$

In Fig. 1, we plot LMC (normalized with respect to the zero-field conductivity) as a function of the magnetic field. The conductivity is quadratic in the magnetic field. When

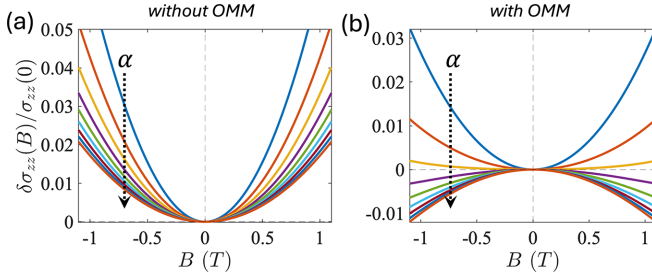


FIG. 1. Longitudinal magnetoconductivity in a pseudospin-1 semimetal with and without including the effect of orbital magnetic moment. Increasing the relative magnitude of the internode scattering strength (α) results in a reversal of the sign of LMC from positive to negative.

$\alpha < \alpha_c^{(1)}$, the conductivity is positive and switches signs when $\alpha > \alpha_c^{(1)}$, where $\alpha_c^{(1)}$ is the critical relative internode scattering strength. When the effects of the orbital magnetic moment are turned off, we only obtain positive LMC irrespective of α . Increasing α only reduces the magnitude of the conductivity but does not change the sign. Before we discuss further, it is useful to compare and contrast our results with a conventional spin-1/2 Weyl semimetal. Son and Spivak [18] predicted positive and quadratic LMC in Weyl semimetals driven by chirality-breaking internode scattering. However, several recent studies later proposed that intranode scattering, by itself, could lead to positive longitudinal magnetoconductivity by including a force term $\mathbf{E} \cdot \mathbf{B}$ in equations of motion [21–27]. However, these results do not account for global charge conservation or orbital magnetic moment and assume a constant relaxation time, both of which are crucial to obtain the correct magnetoconductivity results. Going beyond the standard approximations reveals that LMC in Weyl fermions is negative for zero internode scattering [53], positive for infinitesimal internode scattering, and negative if the internode scattering is large enough [39,53]. The results obtained here for pseudospin-1 fermions also paint a similar picture, but a closer comparison sheds further light on their universal features.

A comparison of LMC between Weyl fermions and pseudospin-1 fermions is made in Fig. 2(a). Note that magnetoconductivity has a larger magnitude for pseudospin-1

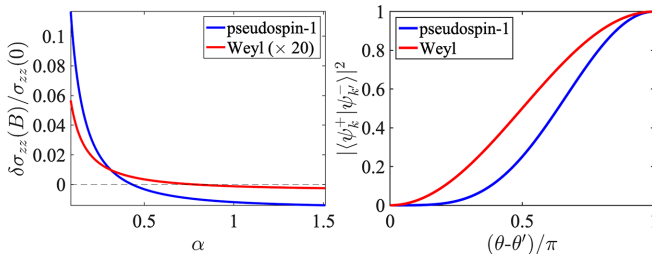


FIG. 2. (a) Longitudinal magnetoconductance for a fixed magnetic field as a function of the intervalley scattering strength α . The switch from positive to negative LMC happens at a higher value for Weyl fermions compared to pseudospin 1. (b) Overlap between fermions at different valleys $|\langle\psi_{\mathbf{k}}^+|\psi_{\mathbf{k}}^-\rangle|^2$ for a fixed azimuthal angle ϕ as a function of the difference of incoming (θ) and outgoing (θ') polar angle.

fermions and switches its sign from positive to negative for a much smaller value of internode scattering strength. We intuitively understand this by first examining the physics at zero magnetic field. We compare the overlap of fermions in opposite valleys ($|\langle\psi_{\mathbf{k}}^+|\psi_{\mathbf{k}}^-\rangle|^2$), which is responsible for chirality breaking transport (chiral anomaly). This is shown in Fig. 2(b). Clearly, pseudospin-1 fermions are more likely to get backscattered while flipping their chirality compared to Weyl fermions. The ratio (l) can be naively estimated as

$$l = \left(\frac{\int_{\pi/2}^{\pi} |\langle\psi_{\mathbf{k}}^+|\psi_{\mathbf{k}}^-\rangle|^2}{\int_0^{\pi/2} |\langle\psi_{\mathbf{k}}^+|\psi_{\mathbf{k}}^-\rangle|^2} \right)_{s=1} \left(\frac{\int_{\pi/2}^{\pi} |\langle\psi_{\mathbf{k}}^+|\psi_{\mathbf{k}}^-\rangle|^2}{\int_0^{\pi/2} |\langle\psi_{\mathbf{k}}^+|\psi_{\mathbf{k}}^-\rangle|^2} \right)_{s=1/2}^{-1} \\ = \frac{(8 + 3\pi)}{(6 + 3\pi)} \sim 1.12, \quad (24)$$

where s refers to the pseudospin. This increased likelihood of backscattering results in a quicker conductivity decrease due to internode scattering. We further analytically evaluate the zero-field conductivity using the Boltzmann formalism to

$$\sigma_{zz}^{s=1} = \frac{e^2 v_F^2}{V \pi^2 (3\alpha + 1)}, \\ \sigma_{zz}^{s=1/2} = \frac{e^2 v_F^2}{16V \pi^2 (2\alpha + 1)}, \quad (25)$$

where $V = nU^2/\hbar$ (U being the strength of the impurity potential). The conductivity for pseudospin-1 fermions is greater in magnitude than Weyl fermions for the same set of parameters, and again, we note that conductivity depends more strongly on internode scattering in pseudospin-1 fermions than in Weyl fermions. In the absence of intervalley scattering, it is straightforward to evaluate that for fermions with higher pseudospin ($s > 1$), the zero-field conductivity also increases with s . We conjecture that the conductivity in the presence of intervalley scattering (α) should also drop more dramatically with increasing s (although this must be confirmed by explicit calculations). This suggests that we may need more diagnostic tools other than relying on negative magnetoresistance studies to confirm chirality-breaking scattering in systems with higher pseudospin- s fermions.

B. Semimetals with fourfold degeneracy

We extend our study to pseudospin-3/2 fermions, which are fourfold degenerate. The Hamiltonian for a fourfold degenerate Weyl point having chirality χ is given by

$$H(\mathbf{k}) = \sum_{\chi=\pm 1} \chi \hbar v_F \mathbf{k} \cdot \mathbf{S}, \quad (26)$$

where $\mathbf{S} = \{S_x, S_y, S_z\}$ are the vector components of the generator $\mathbf{S}$ with spin 3/2. Their exact form is provided in Appendix C. The eigenvalues of the Hamiltonian are given by

$$\epsilon_k = \pm \frac{1}{2} \hbar v_F k, \quad \pm \frac{3}{2} \hbar v_F k. \quad (27)$$

The Berry curvature and the orbital magnetic moment of all four bands of the Hamiltonian is evaluated to be

$$\mathbf{\Omega}_s^x(\mathbf{k}) = -\frac{\chi s \mathbf{k}}{k^3},$$

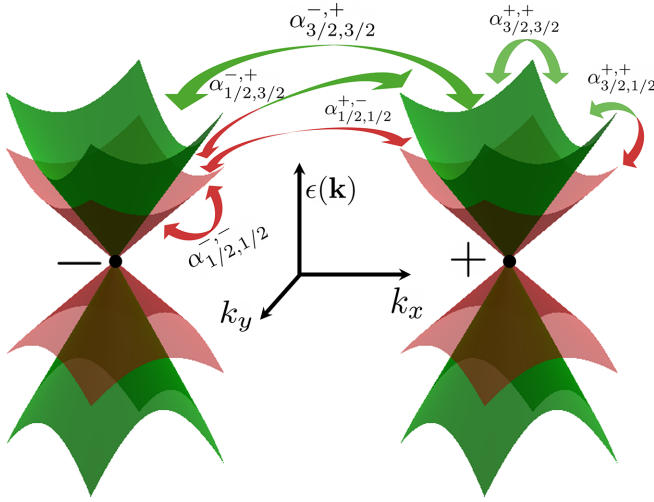


FIG. 3. The band structure of the pseudospin-3/2 fermions at two different nodes/valleys for $k_z = 0$. The different scattering channels has been marked by red and green arrows. There are four qualitatively different types of scattering, i.e., (i) internode interband, (ii) internode intraband, (iii) intranode interband, and (iv) intranode intraband.

$$\mathbf{m}_s^x(\mathbf{k}) = -\frac{e \chi v_F \mathcal{G}_s \mathbf{k}}{k^2}, \quad (28)$$

where the band index $s \in \{\pm\frac{1}{2}, \pm\frac{3}{2}\}$ and corresponding weight factor, $\{\mathcal{G}_{\pm 1/2}, \mathcal{G}_{\pm 3/2}\} = \{\frac{7}{4}, \frac{3}{4}\}$. Without loss of generality, we assume a positive chemical potential, and therefore at each valley we have two Fermi surfaces. This increases the number of scattering channels. We categorize this as (i) internode-interband ($\{\chi, s\} \Leftrightarrow \{\chi' \neq \chi, s' \neq s\}$), (ii) internode-intraband ($\{\chi, s\} \Leftrightarrow \{\chi' \neq \chi, s' = s\}$), (iii) intranode-interband ($\{\chi, s\} \Leftrightarrow \{\chi' = \chi, s' \neq s\}$), and (iv) intranode-intraband ($\{\chi, s\} \Leftrightarrow \{\chi' = \chi, s' = s\}$). In our formalism, the strength of all these scattering channels can be controlled separately and thus their effects can be studied individually. Figure 3 presents the band structure and the different scattering channels. The mathematical and calculation details are relegated to Appendixes C and D. By default, we always have the intraband-intranode scattering present in the system (i.e., there is always scattering between $\{\chi, s\} \Leftrightarrow \{\chi' = \chi, s' = s\}$), and the other scattering

parameters are normalized with respect to this. In other words, $\alpha_{3/2,1/2}^{++} = U_{+\frac{3}{2},+\frac{1}{2}}/U_{+\frac{1}{2},+\frac{1}{2}}$, and so on (also note that we choose $U_{+\frac{1}{2},+\frac{1}{2}} = U_{+\frac{3}{2},+\frac{3}{2}} = U_{-\frac{1}{2},-\frac{1}{2}} = U_{-\frac{3}{2},-\frac{3}{2}}$). Furthermore, unless otherwise mentioned, while studying the effect of one scattering channel, all the other qualitatively different scattering channel strengths (except the intranode-intraband channel that is used for normalization) are set to zero. This allows us to isolate the behavior of qualitatively different scattering mechanisms.

Figure 4(a) presents the behavior of magnetoconductivity as the intranode-interband scattering $\alpha_{3/2,1/2}^{++}$ is varied. Interestingly, we discover that while the magnetoconductivity decreases with increasing $\alpha_{3/2,1/2}^{++}$, there is no sign reversal, and the LMC always stays positive. Note that while we vary the parameter $\alpha_{3/2,1/2}^{++}$, the corresponding parameter on the other node ($\alpha_{3/2,1/2}^{--}$) is also varied simultaneously and the calculated LMC is the sum of both nodes. In Fig. 4(b), we plot the magnetoconductivity as a function of varying the internode-intraband scattering parameter $\alpha_{1/2,1/2}^{+-}$. Here, just like the Weyl and pseudospin-1/2 case, LMC exhibits sign reversal as $\alpha_{1/2,1/2}^{+-}$ is increased. We observe that the change of signs from positive to negative happens at a much smaller value of $\alpha_{1/2,1/2}^{+-}$ compared to the Weyl and pseudospin-1 case—which aligns with our earlier discussion. Again, we note that while we vary the parameter $\alpha_{1/2,1/2}^{+-}$, the corresponding parameter for the other band ($\alpha_{3/2,3/2}^{+-}$) is also varied simultaneously and the calculated LMC is the sum of both bands. Finally, in Fig. 4(c) we examine the effect of internode-interband scattering ($\alpha_{3/2,1/2}^{+-}$), which shows a change of signs from negative to positive. Remarkably, each of the three distinct scattering channels yields a qualitatively different LMC behavior.

V. OUTLOOK

Condensed matter systems provide a unique platform for studying emergent fermions that otherwise have no analogs in high-energy physics. Pseudospin-1 fermions form one such example that can emerge in candidate materials with space group symmetries 199, 214, and 220. Similarly, higher pseudospin excitations are possible as well [28]. Investigating the effect of chirality, Berry curvature and their manifestation in transport experiments can reveal fascinating properties of (pseudo)relativistic fermions beyond the stan-

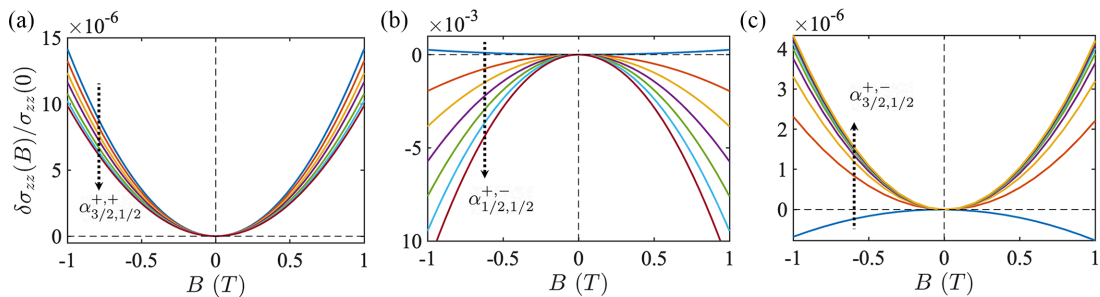


FIG. 4. Longitudinal magnetoconductivity (normalized) for pseudospin-3/2 Weyl fermions. (a) The effect of varying of intranode-interband scattering ($\alpha_{3/2,1/2}^{++}$). (b) Effect of varying internode-intraband scattering ($\alpha_{1/2,1/2}^{+-}$). (c) Effect of varying internode-interband scattering ($\alpha_{3/2,1/2}^{+-}$). In each plot, the particular parameter is increased in the direction of the arrow.

standard Dirac fermions and shed light on the universality of (pseudo)relativistic fermions. While the magnetoconductivity in the presence of internode scattering quickly becomes negative with increasing pseudospin index (1/2 to 1), the magnetoconductivity for higher pseudospin fermions (3/2 and beyond) shows a richer behavior due to the presence of multiple Fermi surfaces at each nodal point. Our study will help design upcoming experiments on candidate materials where such excitations can emerge.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

APPENDIX A: SUBSTITUTION AND EXPRESSION FOR SYSTEM OF LINEAR EQUATIONS

The system of linear equations of the unknown variables involved in Λ^χ is written as follows:

$$\begin{bmatrix} a^+ \\ b^+ \\ c^+ \\ a^- \\ b^- \\ c^- \end{bmatrix} = \begin{bmatrix} \Pi^{++}T^+ & \Pi^{++}I^+ & \Pi^{++}J^+ & \Pi^{+-}I^- & \Pi^{+-}M^- & \Pi^{+-}N^- \\ \Pi^{++}I^+ & \Pi^{++}M^+ & \Pi^{++}N^+ & \Pi^{+-}T^- & \Pi^{+-}I^- & \Pi^{+-}J^- \\ \Pi^{++}J^+ & \Pi^{++}N^+ & \Pi^{++}Q^+ & \Pi^{+-}J^- & \Pi^{+-}N^- & \Pi^{+-}Q^- \\ \Pi^{--}I^+ & \Pi^{--}M^+ & \Pi^{--}N^+ & \Pi^{--}T^- & \Pi^{--}I^- & \Pi^{--}J^- \\ \Pi^{--}T^+ & \Pi^{--}I^+ & \Pi^{--}J^+ & \Pi^{--}I^- & \Pi^{--}M^- & \Pi^{--}N^- \\ \Pi^{--}J^+ & \Pi^{--}N^+ & \Pi^{--}Q^+ & \Pi^{--}J^- & \Pi^{--}N^- & \Pi^{--}Q^- \end{bmatrix} \cdot \begin{bmatrix} a^+ \\ b^+ \\ c^+ \\ a^- \\ b^- \\ c^- \end{bmatrix} - \begin{bmatrix} \Pi^{++}G^+ + \Pi^{+-}L^- \\ \Pi^{++}L^+ + \Pi^{+-}G^- \\ \Pi^{++}P^+/4 + \Pi^{+-}P^-/4 \\ \Pi^{--}L^+ + \Pi^{--}G^- \\ \Pi^{--}G^+ + \Pi^{--}L^- \\ \Pi^{--}P^+/4 + \Pi^{--}P^-/4 \end{bmatrix}. \quad (\text{A1})$$

Substitution in the above matrix equation is written as

$$\begin{aligned} \int d\theta' f^\chi(\theta') \cos^4(\theta'/2) &= R^\chi, & \int d\theta' f^\chi(\theta') \sin^4(\theta'/2) &= S^\chi, \\ \int d\theta' f^\chi(\theta') \sin^8(\theta'/2) &= M^\chi, & \int d\theta' f^\chi(\theta') h^\chi(\theta') &= U^\chi, \\ \int d\theta' f^\chi(\theta') \cos^8(\theta'/2) &= T^\chi, & \int d\theta' f^\chi(\theta') \sin^2(\theta') &= O^\chi, \\ \int d\theta' f^\chi(\theta') h^\chi(\theta') \sin^2(\theta') &= P^\chi, & \int d\theta' f^\chi(\theta') h^\chi(\theta') \sin^4(\theta'/2) &= L^\chi, \\ \int d\theta' f^\chi(\theta') h^\chi(\theta') \cos^4(\theta'/2) &= G^\chi, & \int d\theta' f^\chi(\theta') h^\chi(\theta') \cos^4(\theta'/2) \sin^2(\theta') &= J^\chi, \\ \int d\theta' f^\chi(\theta') h^\chi(\theta') \sin^4(\theta'/2) \sin^2(\theta') &= N^\chi, & \int d\theta' f^\chi(\theta') h^\chi(\theta') \sin^4(\theta'/2) \cos^4(\theta'/2) &= I^\chi, \\ \int d\theta' f^\chi(\theta') \sin^4(\theta') &= Q^\chi. \end{aligned}$$

APPENDIX B: ZERO-FIELD CONDUCTIVITY

We calculate the zero-field conductivity and derive its analytical expressions for two systems, i.e., spin 1/2 and pseudospin 1 using the formalism mentioned in the main text. For $\mathbf{B} = 0$, the coupled Eqs. (8) acquire the following form:

$$\dot{\mathbf{r}}_s^\chi = \frac{e}{\hbar} (\mathbf{E} \times \boldsymbol{\Omega}_s^\chi) + \mathbf{v}_{\mathbf{k}_s^\chi}^\chi, \quad \dot{\mathbf{k}}_s^\chi = -\frac{e}{\hbar} \mathbf{E}. \quad (\text{B1})$$

Using this, the Boltzmann equation is simplified to

$$v_{z,\mathbf{k}}^\chi + \frac{\Lambda^\chi(\theta)}{\tau_\mu^\chi} = \sum_{\chi'} \mathcal{V} \int \frac{d^3\mathbf{k}'}{(2\pi)^3} \mathbf{W}_{\mathbf{k}\mathbf{k}'}^{\chi\chi'} \Lambda^{\chi'}(\theta'). \quad (\text{B2})$$

We solve this analytically using the ansatz defined in the main text and using the ansatz defined in Refs. [41,43] for Weyl (spin 1/2). The net scattering rate at two valleys ($\chi = \pm 1$) for $B = 0$ is calculated to be

$$\frac{1}{\tau_{\mu}^{\chi,s=1}} = \frac{4\pi V \epsilon_F^2 (\alpha + 1)}{3\hbar^3 v_F^3} \equiv \frac{8}{\tau_{\mu}^{\chi,s=1/2}}. \quad (\text{B3})$$

Returning back to Eq. (B2), it is written explicitly in terms of unknown variables involved in ansatz for spin-1 WSMs as

$$\lambda^+ - \frac{\sin^2(\theta)(10\lambda^+ + 3a^+ + 3b^+ - 12c^+ + 10\lambda^- + 3\alpha a^- + 3\alpha b^- - 20\alpha c^+ + 8\alpha c^-)}{20(\alpha + 1)}$$

$$\begin{aligned}
& - \frac{\cos^4\left(\frac{\theta}{2}\right)(10\lambda^+ - 4a^+ + b^+ + 6c^+ - 5v_F + 10\lambda^-\alpha - 10\alpha a^+ + \alpha a^- + 6\alpha b^- + 6\alpha c^- + 5\alpha v_F)}{10(\alpha + 1)} \\
& - \frac{\sin^4\left(\frac{\theta}{2}\right)(10\lambda^+ + a^+ - 4b^+ + 6c^+ + 5v_F + 10\lambda^-\alpha + 6\alpha a^- - 10\alpha b^+ + \alpha b^- + 6\alpha c^- - 5\alpha v_F)}{10(\alpha + 1)} = 0, \quad (\text{B4}) \\
& \lambda^- - \frac{\sin^2(\theta)(10\lambda^- + 3a^- + 3b^- - 12c^- + 10\lambda^+\alpha + 3\alpha a^+ + 3\alpha b^+ + 8\alpha c^+ - 20\alpha c^-)}{20(\alpha + 1)} \\
& - \frac{\cos^4\left(\frac{\theta}{2}\right)(10\lambda^- - 4a^- + b^- + 6c^- - 5v_F + 10\lambda^+\alpha + \alpha a^+ - 10\alpha a^- + 6\alpha b^+ + 6\alpha c^+ + 5\alpha v_F)}{10(\alpha + 1)} \\
& - \frac{\sin^4\left(\frac{\theta}{2}\right)(10\lambda^- + a^- - 4b^- + 6c^- + 5v_F + 10\lambda^+\alpha + 6\alpha a^+ + \alpha b^+ - 10\alpha b^- + 6\alpha c^+ - 5\alpha v_F)}{10(\alpha + 1)} = 0. \quad (\text{B5})
\end{aligned}$$

Equating the coefficient of $\cos^4\left(\frac{\theta}{2}\right)$, $\sin^4\left(\frac{\theta}{2}\right)$ and $\sin^2(\theta)$ in these two equations results in a system of eight coupled equations to be solved for eight unknown variables ($\lambda^\pm, a^\pm, b^\pm, c^\pm$). The solution obtained is as follows: $\lambda^\pm = 0$, $a^\pm = \frac{v_F(\alpha-1)}{3\alpha+1}$, $b^\pm = a^\pm$, $c^\pm = 0$. Using Eqs. (10), (22), and (B3), LMC is evaluated to be

$$\sigma_{zz}^{s=1} = \frac{e^2 v_F^2}{V\pi^2(3\alpha+1)}, \quad \sigma_{zz}^{s=1/2} = \frac{e^2 v_F^2}{16V\pi^2(2\alpha+1)}. \quad (\text{B6})$$

APPENDIX C: SEMIMETALS WITH FOURFOLD DEGENERACY

The Hamiltonian for a fourfold degenerate Weyl point having chirality χ is given by

$$H(\mathbf{k}) = \sum_{\chi=\pm 1} \chi \hbar v_F \mathbf{k} \cdot \mathbf{S}, \quad (\text{C1})$$

where $\mathbf{S} = \{S_x, S_y, S_z\}$ are the vector components of the generator $\mathbf{S}$ with spin 3/2 and have the following explicit form:

$$S_x = \begin{pmatrix} 0 & \frac{\sqrt{3}}{2} & 0 & 0 \\ \frac{\sqrt{3}}{2} & 0 & 1 & 0 \\ 0 & 1 & 0 & \frac{\sqrt{3}}{2} \\ 0 & 0 & \frac{\sqrt{3}}{2} & 0 \end{pmatrix}, \quad S_y = \begin{pmatrix} 0 & \frac{-i\sqrt{3}}{2} & 0 & 0 \\ \frac{i\sqrt{3}}{2} & 0 & -i & 0 \\ 0 & i & 0 & \frac{-i\sqrt{3}}{2} \\ 0 & 0 & \frac{i\sqrt{3}}{2} & 0 \end{pmatrix}, \quad S_z = \begin{pmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{3}{2} \end{pmatrix}. \quad (\text{C2})$$

The energy eigenvalues are given by

$$\epsilon_k = \pm \frac{1}{2} \hbar v_F k, \quad \pm \frac{3}{2} \hbar v_F k. \quad (\text{C3})$$

The Berry curvature and OMM of all four bands of the Hamiltonian is evaluated to be

$$\boldsymbol{\Omega}_s^x(\mathbf{k}) = -\frac{\chi s \mathbf{k}}{k^3}, \quad \mathbf{m}_s^x(\mathbf{k}) = -\frac{e \chi v_F \mathcal{G}_s \mathbf{k}}{k^2}, \quad (\text{C4})$$

where the band index $s \in \{\pm \frac{1}{2}, \pm \frac{3}{2}\}$ and corresponding weight factor $\{\mathcal{G}_{\pm 1/2}, \mathcal{G}_{\pm 3/2}\} = \{\frac{7}{4}, \frac{3}{4}\}$. We categorize the scattering channels as (i) internode interband, (ii) internode intraband, (iii) intranode interband, and (iv) intranode intraband. In our formalism, the strength of all these scattering channels can be controlled separately. The overlap of the wave functions ($\mathcal{T}_{s,s'}^{xx'}$) acquires the following form:

$$\begin{aligned}
\mathcal{T}_{3/2,1/2}^{+,+} &= \frac{1}{8}(3\pi + 3\pi \cos^2 \theta + 3\pi \cos^2 \theta' - 9\pi \cos^2 \theta \cos^2 \theta' - 15\pi \cos^3 \theta \cos^3 \theta' - 3\pi \cos \theta \cos \theta' \\
&\quad + 9\pi \cos \theta \cos^3 \theta' + 9\pi \cos^3 \theta \cos \theta'); \\
\mathcal{T}_{3/2,3/2}^{+,+} &= \frac{1}{8}(5\pi - 3\pi \cos^2 \theta - 3\pi \cos^2 \theta' + 9\pi \cos^2 \theta \cos^2 \theta' + 5\pi \cos^3 \theta \cos^3 \theta' + 9\pi \cos \theta \cos \theta' \\
&\quad - 3\pi \cos \theta \cos^3 \theta' - 3\pi \cos^3 \theta \cos \theta'); \\
\mathcal{T}_{3/2,1/2}^{+,-} &= \frac{1}{8}(3\pi + 3\pi \cos^2 \theta + 3\pi \cos^2 \theta' - 9\pi \cos^2 \theta \cos^2 \theta' + 15\pi \cos^3 \theta \cos^3 \theta' + 3\pi \cos \theta \cos \theta' \\
&\quad - 9\pi \cos \theta \cos^3 \theta' - 9\pi \cos^3 \theta \cos \theta'); \\
\mathcal{T}_{3/2,3/2}^{+,-} &= \frac{1}{8}(5\pi - 3\pi \cos^2 \theta - 3\pi \cos^2 \theta' + 9\pi \cos^2 \theta \cos^2 \theta' - 5\pi \cos^3 \theta \cos^3 \theta' - 9\pi \cos \theta \cos \theta' \\
&\quad + 3\pi \cos \theta \cos^3 \theta' + 3\pi \cos^3 \theta \cos \theta'); \\
\mathcal{T}_{1/2,1/2}^{+,+} &= \frac{1}{8}(5\pi - 3\pi \cos^2 \theta - 3\pi \cos^2 \theta' + 9\pi \cos^2 \theta \cos^2 \theta' + 45\pi \cos^3 \theta \cos^3 \theta' + 17\pi \cos \theta \cos \theta' \\
&\quad - 27\pi \cos \theta \cos^3 \theta' - 27\pi \cos^3 \theta \cos \theta');
\end{aligned}$$

$$\begin{aligned}
\mathcal{T}_{1/2,1/2}^{+,-} &= \frac{1}{8}(5\pi - 3\pi \cos^2 \theta - 3\pi \cos^2 \theta' + 9\pi \cos^2 \theta \cos^2 \theta' - 45\pi \cos^3 \theta \cos^3 \theta' - 17\pi \cos \theta \cos \theta' \\
&\quad + 27\pi \cos \theta \cos^3 \theta' + 27\pi \cos^3 \theta \cos \theta'); \\
\mathcal{T}_{1/2,3/2}^{+,+} &= \mathcal{T}_{3/2,1/2}^{+,+}; \quad \mathcal{T}_{3/2,1/2}^{-,-} = \mathcal{T}_{3/2,1/2}^{+,+}; \quad \mathcal{T}_{1/2,3/2}^{-,-} = \mathcal{T}_{3/2,1/2}^{-,-}; \quad \mathcal{T}_{1/2,3/2}^{-,+} = \mathcal{T}_{3/2,1/2}^{+,-}; \quad \mathcal{T}_{3/2,1/2}^{-,+} = \mathcal{T}_{3/2,1/2}^{+,-}; \\
\mathcal{T}_{1/2,3/2}^{+,-} &= \mathcal{T}_{3/2,1/2}^{+,-}; \quad \mathcal{T}_{3/2,3/2}^{+,+} = \mathcal{T}_{3/2,3/2}^{+,-}; \quad \mathcal{T}_{1/2,1/2}^{-,+} = \mathcal{T}_{1/2,1/2}^{+,-}; \quad \mathcal{T}_{3/2,3/2}^{-,-} = \mathcal{T}_{3/2,3/2}^{+,-}; \quad \mathcal{T}_{1/2,1/2}^{-,-} = \mathcal{T}_{1/2,1/2}^{+,+}.
\end{aligned}$$

APPENDIX D: BOLTZMANN TRANSPORT FOR A SYSTEM WITH MULTIPLE BANDS

For a system with multiple bands characterized by different set of chiralities (χ) and band index (s), the distribution function in each band is denoted by $f_{\mathbf{k}}^m$, where $m \in \{(\chi, s) : \chi = \pm 1 \text{ and } s = \pm 1/2, \pm 3/2\}$. Generalizing the formalism presented in the main text, the collision integral must take into account scattering between multiple bands that may or may not be of the same chirality χ and/or band index s , i.e., $m = (\chi, s) \Leftrightarrow (\chi', s') = p$. Thus, $\mathcal{I}_{\text{col}}[f_{\mathbf{k}}^m]$ can be expressed as

$$\mathcal{I}_{\text{col}}[f_{\mathbf{k}}^m] = \sum_p \sum_{\mathbf{k}'} W_{\mathbf{k},\mathbf{k}'}^{mp} (f_{\mathbf{k}'}^p - f_{\mathbf{k}}^m), \quad (\text{D1})$$

where the sum p runs over all possible combination of χ and s (as defined in above), and scattering rate $W_{\mathbf{k},\mathbf{k}'}^{mp}$ in the first Born approximation is given by

$$W_{\mathbf{k},\mathbf{k}'}^{mp} = \frac{2\pi}{\hbar} \frac{n}{V} |\langle \psi_{\mathbf{k}'}^p | U_{\mathbf{k}\mathbf{k}'}^{mp} | \psi_{\mathbf{k}}^m \rangle|^2 \delta(\epsilon_{\mathbf{k}'}^p - \epsilon_{\mathbf{k}}^m). \quad (\text{D2})$$

The scattering potential profile $U_{\mathbf{k}\mathbf{k}'}^{mp}$ can be chosen such that scattering between all the bands is considered appropriately. Proceeding as before, we define the band scattering time $\tau_{\mathbf{k}}^m$ as

$$\frac{1}{\tau_{\mu}^m(\theta)} = \nu \sum_p \iint \frac{\beta^{mp}(k'^p)^3}{|\mathbf{v}_{\mathbf{k}'}^p \cdot \mathbf{k}'^p|} \sin \theta' \mathcal{G}^{mp}(\mathcal{D}_{\mathbf{k}'}^p)^{-1} d\theta', \quad (\text{D3})$$

and the Boltzmann equation becomes

$$h_{\mu}^m(\theta) + \frac{\Lambda_{\mu}^m(\theta)}{\tau_{\mu}^m(\theta)} = \nu \sum_p \iint \frac{\beta^{mp}(k')^3}{|\mathbf{v}_{\mathbf{k}'}^p \cdot \mathbf{k}'^p|} \sin \theta' \mathcal{T}^{mp}(\mathcal{D}_{\mathbf{k}'}^p)^{-1} \Lambda_{\mu}^p(\theta') d\theta'. \quad (\text{D4})$$

Choosing the ansatz $\Lambda_{\mu}^m(\theta) = (\lambda^m - h_{\mu}^m(\theta) + a^m \cos \theta + b^m \cos^2 \theta + c^m \cos^3 \theta) \tau_{\mu}^m(\theta)$ and using the particle number conservation constraint, the Boltzmann equation is reduced to a system of $4m$ equations to be solved for $4m$ unknowns ($m = 2 \times$ the number of positive bands in one valley). Even though, in the current work, we have taken equal numbers of bands in a both valleys, this formalism can be used for the system with an unequal number of bands in valleys having opposite chiralities, i.e., systems with different degeneracies of the Weyl node at different valleys.

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